ITA06020 MACHINE LEARNING

Experiment 13 – Car Price Prediction Using Python

Aim:

To implement a Car Price Prediction model using a suitable Machine Learning algorithm in Python to predict the selling price of a car based on its features.

Dataset: Automobile Dataset (UCI Machine Learning Repository).

Algorithm: Linear Regression.

Procedure

1. Import the required Python libraries.
2. Load the Automobile dataset.
3. Select relevant car features and selling price.
4. Remove missing values.
5. Convert categorical features into numerical values.
6. Split the dataset into training and testing data.
7. Create and train a Linear Regression model.
8. Predict car prices and evaluate the model using R^2 score and Mean Squared Error.

Python Program

```
import pandas as pd

from sklearn.model_selection import train_test_split

from sklearn.linear_model import LinearRegression

from sklearn.metrics import mean_squared_error, r2_score

url = "https://archive.ics.uci.edu/ml/machine-learning-databases/autos/imports-85.data"

columns = [

    'symboling', 'normalized-losses', 'make', 'fuel-type',

    'aspiration', 'num-doors', 'body-style', 'drive-wheels',

    'engine-location', 'wheel-base', 'length', 'width',

    'height', 'curb-weight', 'engine-type', 'num-cylinders',

    'engine-size', 'fuel-system', 'bore', 'stroke',
```

```
    'compression-ratio', 'horsepower', 'peak-rpm',
    'city-mpg', 'highway-mpg', 'price'
]
df = pd.read_csv(url, names=columns, na_values='?')
df = df[['engine-size', 'horsepower', 'curb-weight',
         'highway-mpg', 'price']]
df = df.dropna()
df = df.astype(float)
X = df.drop('price', axis=1)
y = df['price']
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
model = LinearRegression()
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
print("Car Price Prediction")
print("Actual Price:", y_test.iloc[0])
print("Predicted Price:", round(y_pred[0], 2))
print("MSE:", round(mean_squared_error(y_test, y_pred), 2))
print("R2 Score:", round(r2_score(y_test, y_pred), 2))
```

Output:

```
Car Price Prediction
Actual Price: 6989.0
Predicted Price: 11246.83
MSE: 25126979.11
R2 Score: 0.78
```

Result

Thus, the Linear Regression algorithm was successfully implemented using Python and the Automobile dataset to predict car prices based on different car features.